

Skill-4

1. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
ananya@DESKTOP-42I5F93:~$ mkdir skill4
ananya@DESKTOP-42I5F93:~$ cd skill4
ananya@DESKTOP-42I5F93:~/skill4$ pwd
/home/ananya/skill4
ananya@DESKTOP-42I5F93:~/skill4$ nano lexer.c
ananya@DESKTOP-42I5F93:~/skill4$ gcc lexer.c -o lexer
ananya@DESKTOP-42I5F93:~/skill4$ ls
lexer  lexer.c
ananya@DESKTOP-42I5F93:~/skill4$ ./lexer
Enter command: ls -l /home

Tokens:
WORD: ls
WORD: -l
WORD: /home
```

```

ananya@DESKTOP-42I5F93:~/skill4$ ls -l | grep txt
ananya@DESKTOP-42I5F93:~/skill4$ touch text.txt
ananya@DESKTOP-42I5F93:~/skill4$ ls -l | grep txt
-rw-r--r-- 1 ananya ananya 0 Sep 18 03:16 text.txt
ananya@DESKTOP-42I5F93:~/skill4$ ls -l | grep txt
-rw-r--r-- 1 ananya ananya 0 Sep 18 03:16 text.txt
ananya@DESKTOP-42I5F93:~/skill4$ nano parser.c
ananya@DESKTOP-42I5F93:~/skill4$ gcc parser.c -o parser
ananya@DESKTOP-42I5F93:~/skill4$ ls
lexer lexer.c parser parser.c text.txt
ananya@DESKTOP-42I5F93:~/skill4$ ./parser
Enter command: ls -l
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
ananya@DESKTOP-42I5F93:~/skill4$ ./parser
Enter command: ls -l | grep txt
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
ananya@DESKTOP-42I5F93:~/skill4$ ./parser
Enter command: ls -l | grep txt
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
ananya@DESKTOP-42I5F93:~/skill4$ ./parser
Enter command: | ls
Syntax Error: command expected before |
ananya@DESKTOP-42I5F93:~/skill4$ ls -l
total 40
-rwxr-xr-x 1 ananya ananya 16176 Sep 18 03:15 lexer
-rw-r--r-- 1 ananya ananya 813 Sep 18 03:14 lexer.c
-rwxr-xr-x 1 ananya ananya 16176 Sep 18 03:18 parser
-rw-r--r-- 1 ananya ananya 1057 Sep 18 03:17 parser.c
-rw-r--r-- 1 ananya ananya 0 Sep 18 03:16 text.txt
ananya@DESKTOP-42I5F93:~/skill4$ ./lexer
Enter command: ls -l | grep txt

Tokens:
WORD: ls
WORD: -l
DELIMITER: |
WORD: grep
WORD: txt

```

```

ananya@DESKTOP-42I5F93:~/skill4$ ./parser
Enter command: ls -l | grep txt
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
ananya@DESKTOP-42I5F93:~/skill4$ ./parser
Enter command: | ls
Syntax Error: command expected before |

```

